Naimiris A Profit-Driven Smart Farming Model for Scotch Bonnet Peppers

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Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

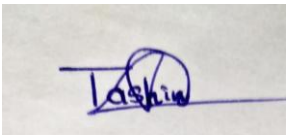
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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Name	Date	Signature

ABSTRACT

Agriculture is one of the key sectors of the Sri Lankan economy. The cultivation of Nai Miris (Scotch Bonnet pepper) is one of the main crops grown by farmers in the wet zone of Sri Lanka. The problem of pest attack is one of the main issues that affect crop growth and productivity. Farmers find it difficult to identify pest attack at an early stage and use chemicals without proper guidance. The existing system of plant disease detection mainly focuses on identifying the infected plant but does not make recommendations on how to treat the plant and the amount of pesticide to be used according to the severity of the pest attack.

In order to resolve the above problem, the present research introduces the concept of an AI-Driven Pest Monitoring Control System, which assists farmers in the detection of pest infestations and the level of severity using the image of leaves obtained through the camera of a smartphone. In the present system, image processing techniques are used to analyze the visual characteristics of the leaves of plants, such as the presence of pests, damage, discoloration, and the spread of the pest. Using the above techniques, the system recognizes the type of pest affecting the Nai Miris plants as well as the level of severity, which can be low, medium, or high.

After the pest and the level of severity have been identified, the system will provide the farmer with automated treatment recommendations, which include the correct dosage of pesticide to be used for the control of the pest. This will help the farmer make timely decisions and use the right amount of chemicals instead of guessing.

The outcome of this study is expected to be a smart and farmer-friendly tool for monitoring pests, which can improve crop protection, minimize pesticide misuse, and enhance agricultural productivity. The system has commercial value as an agricultural solution for digital farming, which can be beneficial for sustainable agriculture in Sri Lanka.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all who supported me in the preparation of this research proposal.Firstly, I would like to thank my research supervisor for their valuable guidance, advice, and motivation in the development of this research project. Their support has helped me to gain a better understanding of the research problem and improve the quality of this research proposal.I would also like to thank all lecturers from the Department of Information Technology, Sri Lanka Institute of Information Technology, for their support in providing necessary knowledge to conduct this research.Moreover, I would like to thank all my friends and colleagues who supported me in the preparation of this research proposal.Last but not least, I would like to express my deepest gratitude to my family for their continuous support, motivation, and encouragement in my academic pursuit.Their support is invaluable in helping me to complete this research proposal successfully.

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LIST OF ABBREVIATIONS

AI - Artificial Intelligence

CNN - Convolutional Neural Network

CV - Computer Vision

DL - Deep Learning

IoT - Internet of Things

YOLO - You Only Look Once (Object Detection Model)

mAP - Mean Average Precision

GPU - Graphics Processing Unit

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Appendix B	Literature Review Comparison Table
Appendix C	Work Breakdown Structure (WBS)

1. INTRODUCTION

This proposal outlines the development of an AI-driven pest monitoring and control system tailored to Nai Miris (Scotch Bonnet chili) cultivation. I begin by situating Nai Miris within the broader context of tropical pepper farming, highlighting the significant yield losses due to pests and diseases in pepper crops [1]. The system's aim is to automatically detect pests and quantify infestation severity via computer vision, then recommend precise pesticide treatments and dosages to minimize crop damage and reduce chemical overuse [2] [3]. Key objectives include designing and training deep learning (DL) models for chili pest/disease classification and severity estimation, building a representative image dataset for Nai Miris, and deploying the models on IoT/edge hardware suitable for smallholder farms [4]. The literature review synthesizes work on DL-based pest detection (CNNs, transformers, segmentation) and severity quantification [5] [6], existing pesticide recommendation methods [7], and IoT/edge deployment strategies. It highlights that while many studies focus on pest classification, few integrate severity assessment or dosage guidance, especially for pepper crops or in Sri Lanka's context. I will close by identifying the research gaps (e.g. lack of pepper-specific AI tools and actionable control recommendations) and arguing the proposed system's impact: improving Nai Miris yields, reducing pesticide misuse, and advancing precision agriculture in the tropical small-farm setting

1.1 Background and Context

Pepper (Capsicum spp.) is a high-value crop globally and in Sri Lanka; Scotch Bonnet ("Nai Miris") chilies are widely grown in the country and abroad. However, chili yields are heavily impacted by pests and diseases. Plant pathogens and insects can reduce global crop productivity by roughly 30–40%. For example, viral, fungal and bacterial diseases in chili often cause premature defoliation and stunting, leading to substantial losses [8]. Similarly, insect pests (such as aphids, thrips, armyworms and fruit borers) infest pepper fields, further diminishing fruit quality and quantity. Traditionally, growers rely on manual scouting and uniform pesticide applications, which is laborious, error-prone, and typically excessive. Sri Lankan chili farmers often lack precise guidance on safe pesticide use; overuse of agrochemicals exacerbates environmental harm and raises costs. By contrast, advances in computer vision (CV) and artificial intelligence (AI) now enable automated, field-based crop monitoring. Convolutional neural networks (CNNs) and other DL models have revolutionized plant disease/pest detection, learning complex visual patterns from large image datasets. Small, affordable IoT devices and.

mobile units (e.g. Raspberry Pi, ESP32 cameras) can run such models on-site. In this context, an integrated system for Nai Miris cultivation – combining AI-powered pest detection, infestation severity quantification, and dosage recommendations – promises to modernize chili farming by providing real-time, data-driven decision support and reducing pesticide reliance.

1.2 Literature Review

AI for Pest Detection in Crops: Computer vision techniques have been widely applied to detect crop pests and diseases. Early methods relied on traditional ML or thresholding, but recent works overwhelmingly use deep CNNs. Region-based models like Faster R-CNN and Mask R-CNN have been successful in instance detection/segmentation [9]. For example, Lin et al. used Faster R-CNN and Mask R-CNN to identify pests on sweet pepper leaves, achieving 89% accuracy with Faster R-CNN and 81% with Mask R-CNN [10]. One-stage detectors like YOLO have become very popular for their speed; A gusting et al. fine-tuned YOLOv5 on an Indonesian red chili dataset (4994 images, 4 pest classes) and reported mAP≈81.3% [11]. More advanced YOLO variants continue to push performance: Wang et al. (2025) introduced YOLO-Pepper (based on YOLOv10) for greenhouse pepper pests/diseases, achieving 94.26% mAP@0.5 on 8046 annotated pepper images [12]. Vision transformers have also been explored for pests; Saranya et al. (2024) proposed a transformer with Hybrid Pooled Multihead Attention, attaining 98% accuracy on a 10-class pest dataset [3]. In general, deep learning-based pest detectors (CNNs, transformers) significantly outperform classic methods by learning robust visual features [13].

Image Datasets: Public datasets such as PlantVillage (disease images) and IP102 (102 classes of insect pests) are often used for model training. Guo et al. (2024) survey noted many works leverage these large datasets [14]. However, pest appearance and environmental context can vary by region/crop, so assembling crop-specific datasets is common. For chili peppers, Leite et al. created a dataset of 1645 diseased chili leaves (Cercospora leaf spot) with pixel-level annotations for severity [15]. Chung et al. (2026) combined public datasets with new field images to train their chili-pepper detection pipeline. In summary, prior studies either use standardized benchmark sets or curate new images for their target crop, often applying data augmentation to increase variety.

Severity Assessment: Quantifying how badly a plant is affected is crucial for recommending treatment. Some studies perform segmentation of lesions (e.g. Mask R-CNN) to compute lesion-to-leaf area ratio as a severity metric. Leite et al. compared Mask R-CNN vs. YOLOv8 for chili leaf spot severity: the Mask R-CNN gave more precise segmentation masks, while YOLOv8 was faster and better at classifying severity stages [15] [5]. Other works classify severity levels directly: Megersa et al. built a dataset of 5000+ maize leaves with common rust symptoms, labeled into 5 severity classes. They trained a ResNet50 classifier to predict severity, achieving ~97.4% accuracy [16] [7]. Image processing methods have also been used: Tewari et al. applied chromatic segmentation on rice leaves to estimate disease spread and control a sprayer valve accordingly [8]. While many papers address disease classification, relatively few explicitly address severity quantification, and almost none tie this to actionable dosing advice.

Pesticide Recommendation and Dosage: A central novelty of this proposal is linking detection to treatment recommendation. Few studies incorporate this. Megersa et al. (2024) is an exception: after classifying rust severity on maize leaves, they translated severity into recommended pesticide dosages, validating their suggestions with plant pathology experts [2]. In precision agriculture, researchers have also developed variable-rate spraying systems: Tewari et al. (2020) built a camera-equipped sprayer that estimates disease area in rice and adjusts pesticide output via solenoid valves, cutting chemical use by ~33.9% [8].These demonstrate the feasibility and value of severity-based dosing. Surveys also note ML-driven models for pesticide recommendation are an emerging field, though practical systems are scarce [2] [3]. In summary, existing work either stops at detection or severity scoring; integrating a decision support for treatment is largely unexplored, especially for chili.

IoT / Edge Deployment: For practical adoption, real-time, on-site processing is important. Cloud-based systems suffer from connectivity issues and latency. Recent work shows that lightweight models on edge devices can maintain high accuracy. Nyakuri et al. (2025) developed a portable IoT device running a tiny CNN (Tiny-LiteNet) on a Raspberry Pi for cereal crop pest/disease detection, achieving 98.6% accuracy while keeping the model size to ~1.2 MB [10]. Chung et al. (2026) similarly designed an edge pipeline for open-field chili: ESP32-CAM nodes feed images to a Pi5 server running a two-stage vision model (YOLOv8n for leaf detection + ResNet-18 for lesion classification) [9] [17]. They demonstrated sub-second end-to-end latency (0.86 s) and robust offline operation. These examples show that mobile and embedded AI can effectively monitor crop health in situ. Combining such

Year	Methods (Model)	Dataset (Size/Source)	Severity Estimation	Treatment/ Dosage Recommendation	Accuracy / Metrics	imitations
2020	Faster R-CNN, Mask R-CNN	Sweet pepper leaves (size N/A)	No	No	89% (Faster R-CNN), 81% (Mask R-CNN)	Small data; no dosage guidance
2020	Image segmentation + VRA	Field rice images (disease/no-disease regions)	Yes (area-based)	No	33.9% chemical use reduction	Tested in rice only; simple segmentation
2023	YOLOv5	Indonesian red chili (4 classes, 4994 imgs)	No	No	mAP@0.5 ≈ 81.3%	No severity/dosage; specific region
2024	Vision Transformer (HPMA)	Novel pest dataset (10 classes)	No	No	98% accuracy	Dataset small; no localization / dosage
2024	ResNet-50	Maize rust severity (5000+ images)	Yes (5 levels)	No	97.4% severity accuracy	Maize-specific; requires large dataset
2025	YOLO-Pepper	Greenhouse pepper pests (8046 images)	No	No	mAP@0.5 = 94.26% @ 115 FPS	Greenhouse only; no dosage component
2025	Mask R-CNN	Chili leaf spot (1645 images)	Yes (segmentation)	No	F1-score up to 0.87	Focused on one disease; small dataset

2026	YOLOv8n + ResNet-18	Open-field chili pepper (public + field images)	Yes (lesion detect)	No	mAP50 ≈ 90%, Acc ≈ 90% (ResNet)	Proof-of-concept; no integrated sprayer logic

architectures with domain-specific models (e.g., pepper pests) is a promising direction.

Crop Specific / Localization: Although many algorithms are generic, crop-specific performance and local adaptation matter. A handful of chili/pepper studies exist. Nurokhman et al. (2024) developed a CNN-based system to detect various chili diseases/pests, reporting very high accuracy [18]. Leite et al. (2025) specifically tackled Cercospora on chili. Similarly, the smartphone system by Tzovaras & Pechlivani (2024) focused on tomato and olive, but demonstrated detection of the tomato pest Tuta absoluta with ~70% mAP [6]. In Sri Lankan context, De Costa et al. (2021) highlight gaps in chili pest management and pesticide knowledge among local farmers [4]. No published system directly addresses Nai Miris in Sri Lanka. This underscores the need for localized data and models. In sum, prior work provides powerful tools (CNN architectures, mobile IoT, etc.), but adaptation to Nai Miris – including local pest species, images, and farmer practices – has not been done.

Table 1 - literature review comparison table

1.3 Research Gap

The reviewed studies collectively underline the promise of AI for crop pest detection, but also reveal clear gaps. Notably, no existing work combines pepper-specific pest detection, infestation quantification, and pesticide decision-making in a unified system. Many studies detect pests with high accuracy [4] [9], but few assess severity or link to control actions [7] [2]. Those that do (e.g. Megersa et al.) are on different crops (maize). Similarly, while edge-AI prototypes exist for corn or vegetables [4], these have not targeted open-field pepper in a tropical small-farm context. In Sri Lanka specifically, there is scant literature on AI for Nai Miris; local research has focused on farmer practices [7] or general plant health, not automated vision systems. Additionally, existing datasets lack Nai Miris-specific pest images; models trained elsewhere may perform poorly on local varieties and illumination conditions. In sum, the gap is an integrated, field-deployable AIoT system for Scotch Bonnet chili that provides end-to-end monitoring: detection, severity scoring, and tailored treatment recommendation.

Addressing this requires assembling new data, developing customized DL models, and engineering for edge deployment – objectives that this proposal tackles directly.

I anticipate several key impacts from this work. Agriculturally, the system will empower Nai Miris farmers with actionable intelligence: early pest alerts and optimal pesticide dosages can significantly reduce crop losses and input costs. For example, adaptive spray systems have shown >30% pesticide savings [2], and expert-validated dosage guidance can prevent over-application. Economically, higher yields and reduced chemical expenses will improve smallholder incomes. Environmentally, minimizing unnecessary pesticide use supports sustainability and food safety goals, in line with Sri Lanka's agricultural policy. Technologically, this project will deliver a novel dataset of Nai Miris pest images and a field-hardened edge-AI pipeline – contributions that can be generalized to other tropical crops. Furthermore, by involving local farmers and extension agents in testing, the project promotes technology adoption and builds capacity in precision agriculture. In sum, this research is significant for its potential to enhance food security and sustainability in Sri Lanka through cutting-edge AI and IoT, addressing an important gap at the intersection of computer vision and crop protection [2].

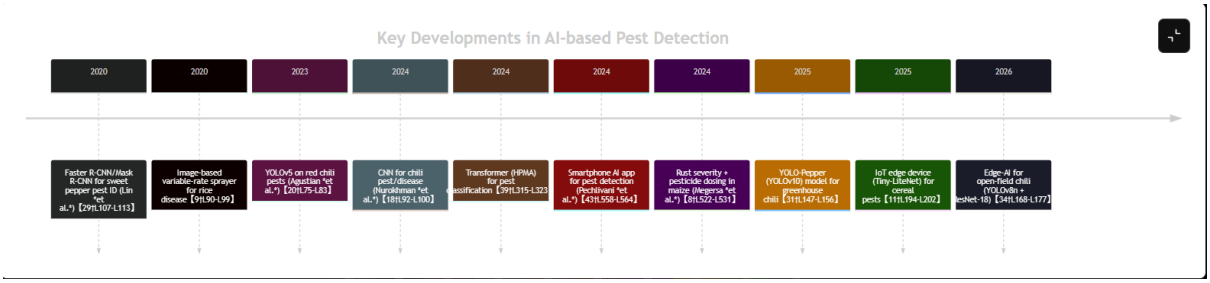


Figure 1 - Research gap

2. OBJECTIVE

2.1 Main Objective

The overall aim of the research is the development of a pest monitoring and control system based on AI, which has the ability to automatically identify pest infestations on Nai Miris plants based on the leaves of the plants, determine the level of the pest infestation, and provide the required pesticide treatment.

2.2 Specific Objectives

1. To create a dataset for Nai Miris leaf images infested by pests in different environmental conditions for machine learning algorithm training and testing.

2. To design and implement image processing techniques for recognizing pest-related image features such as damage, color change, and pest infestation.

3. To implement a pest detection system using deep learning techniques such as YOLOv8 or CNN-based architecture for accurate detection of pest types infesting Nai Miris plants.

4. To implement a pest severity estimation system for classifying pest infestation levels as low, medium, and high based on the extent of damage identified in the image.

5. To implement a decision support system for recommending types and quantities of pesticides required for infested Nai Miris plants based on pest types and infestation levels.

6. To assess the performance of the system using metrics such as accuracy, precision, recall, F1 score, and mean Average Precision (mAP).

3. METHODOLOGY

The research process follows the Design Science Research (DSR) approach, which includes experimental machine learning.

Design Science Research emphasizes the development of the innovative technological artifact, which solves the real-world problem. In the current research, the technological artifact is the AI-based pest monitoring system, which aims at supporting the farmers in the pest control process.

The research process includes the following steps:

1. Problem identification

2. Data collection and dataset development

3. Model development

4. System implementation

5. Performance evaluation

System Development Process

The proposed system has the potential to be created through the following stages.

1. Data Collection

A set of data, i.e., Nai Miris leaves, will be collected from the following sources:

Agricultural farms of Sri Lanka

Public agricultural data

Online agricultural image repositories

The data set will contain the following types of leaves:

Healthy leaves

Pest-infected leaves

Different levels of pest damage

All the collected images will be labeled manually based on the type of pest and severity level.

2. Data Preprocessing

The dataset will go through some preprocessing steps before the machine learning models are trained. These steps include:

Image resizing and normalization

Noise removal

Data augmentation, including rotation, flipping, and brightness

3. Pest Detection Model

A deep learning-based object detection model like YOLOv8 will be used to detect pest infestations on the leaves.

YOLO-based models are appropriate for the task because they offer:

High accuracy levels

Real-time object detection

Efficient execution on mobile devices

The model will be used to detect:

Presence of pests

Location of the pests on the leaves

Damage inflicted by the pests

4. Severity Estimation

After the detection of the pest, the system will be used to estimate the severity of the pest damage using a severity estimation model based on Convolutional Neural Networks (CNN) or image segmentation.

The system will classify the levels of severity based on the damaged area ratio of the leaf as follows:

Low Severity

Medium Severity

High Severity

5.Treatment Recommendation Module

A decision support system will be created, which provides suggestions on pesticide treatment based on the following criteria:

Type of pest

Severity level of the pest

The recommendation system of the decision support system will be based on expert agricultural guidelines, which provide the following suggestions:

Type of pesticide

Dosage level of the pesticide

This system assists farmers in the effective usage of pesticides, reducing the unnecessary use of chemical sprays.

6. System Implementation

The proposed system is to be implemented as a mobile-supported application/web-based system where farmers can:

Capture the image of the leaf of the Nai Miris plant

Upload the image to the system

Access the results of the pest detection

Access the analysis of the severity level

Access the recommendations of the pesticide treatment

Tools and Technologies

The following technologies will be used for system development:

Component Technology

Programming Language Python

Deep Learning Framework TensorFlow / PyTorch

Object Detection Model	YOLOv8
Image Processing	OpenCV
Dataset Annotation	LabelImg
Model Training	Google Colab / GPU workstation
Application Interface	Mobile or Web Application

Data Collection and Ethical Considerations

All image data used in this research is obtained through ethical means and is used for purely academic purposes. Farmers who help in the collection of data in the fields will be made aware of the purpose of the research. There is no collection of personal information during the data acquisition.

System Validation and Performance Evaluation

The performance of the proposed system will be evaluated using several metrics commonly used in machine learning.

These include:

Accuracy – overall prediction correctness

Precision – proportion of correct positive predictions

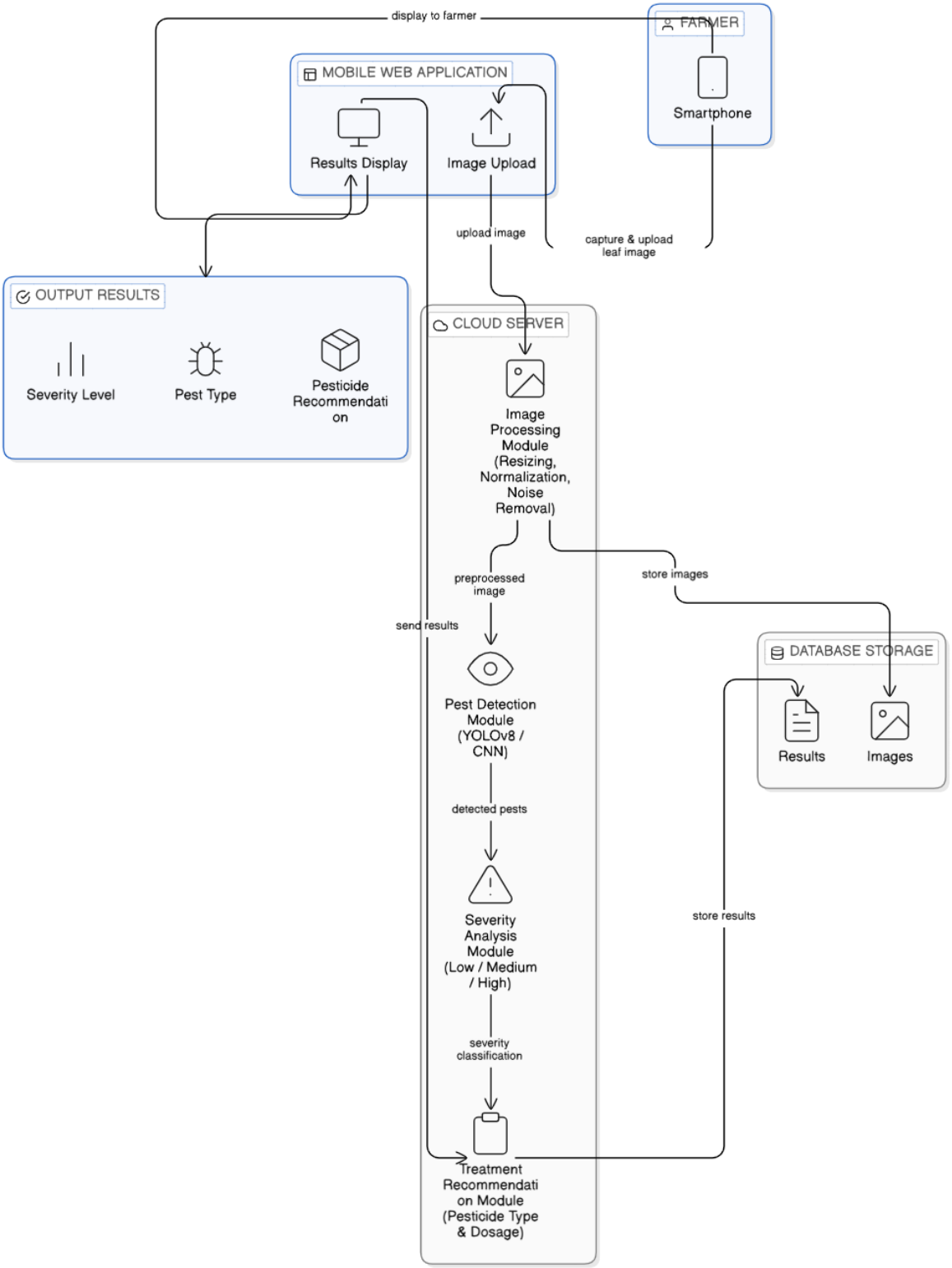
Recall – ability to detect all pest instances

F1-Score – balance between precision and recall

Mean Average Precision (mAP) – object detection accuracy

The trained models will be tested using a separate test dataset to ensure reliable evaluation of system performance.

4. HIGH-LEVEL SYSTEM ARCHITECTURE



The proposed system, AI-Driven Pest Monitoring and Control System, is expected to aid farmers in identifying pest infestation in Nai Miris (Scotch Bonnet pepper) plant species, infestation severity, and provide appropriate advice on the type of pesticide to be used. The system comprises a number of components, all of which interact with each other to ensure intelligent pest management decision-making based on the plant's leaf images.

Image Capture Module

The process starts with the image capture module, where the farmer uses the camera of the smartphone to click images of the Nai Miris plant leaves that are likely to be infested with pests. The images are then sent to the system through a mobile/web application. The image capture module is the first point of input in the entire process and is considered the primary input point in the entire process.

Image Processing Module

Once this image is uploaded, it is then sent to the image processing module, which is present

Figure 2 - system diagram

in the cloud server. This module performs various preprocessing activities, such as resizing, normalizing, removing noise, and enhancing the image, etc. This ensures that the images obtained through capture are suitable for analysis by machine learning models.

Pest Detection Module

After the preprocessing task is done, the processed images are then passed through a deep learning-based pest detection model like YOLOv8 or a Convolutional Neural Network (CNN). In this module, the pest-affected regions on the leaf surface are identified, as well as the specific pest or damage pattern present in the image. In addition, the pest-affected regions are localized, and the required visual features are extracted.

Pest Severity Analysis Module

Once the areas of pest attack have been identified, the system proceeds to the severity analysis. During this stage of the system, the severity of the pest attack is analyzed. The severity of the attack is analyzed on the basis of the extent of the damaged area of the leaf. The severity is classified into different levels such as low, medium, and high levels of infestation.

Treatment Recommendation Module

Upon identification of the pest type and severity level, the system activates the recommendation for treatment. This part includes recommendations for types of pesticides and levels of dosages required for effective control of pests. Recommendations are made using predefined guidelines for agricultural activities. This helps farmers apply pesticides in the right manner and avoid using excess chemicals.

Cloud Server and Data Storage

The AI models, as well as the data processing modules, are housed on a cloud-based server. The cloud-based system allows the system to process large image datasets, compute complex models, as well as store pest detection results for future analysis. Additionally, the cloud-based system allows the system to scale as well as provide access to multiple users.

Component Interaction and Data Flow

The data flow of the system can be described as follows: the farmer takes a picture of the leaves of the plants using the smartphone's camera. The image is then uploaded to the cloud server, where the pre-processing and pest detection algorithms process the image. The detected pest details are then processed by the severity analysis module. The treatment recommendation module then generates the pesticide dosage recommendations and sends the results back to the farmer.

Scalability and Security Considerations

The system is intended to accommodate a number of users through the use of cloud-based technology, which provides storage and processing capabilities in line with the increase in users. Various security measures such as data transmission, user authentication, and data storage will be implemented to protect user data and enhance efficient system operation.

5. USER REQUIREMENTS

Stakeholder Identification

Several stakeholders are involved in the proposed AI-driven pest monitoring system.

Farmers (Primary Users)

The main users of the system are Nai Miris farmers in Sri Lanka's wet-zone regions. They will use the mobile application to capture leaf images and receive pest detection results and pesticide treatment recommendations.

Agricultural Experts

Agricultural officers and pest management specialists can use the system to monitor pest outbreaks and provide improved pest control advice.

Small Agricultural Investors

Individuals investing in small-scale farming operations can use the system to monitor crop health and improve productivity.

Researchers and Developers

Researchers and system developers will use the system for improving AI models, updating pest datasets, and enhancing system performance.

Functional Requirements

The functional requirements describe what the system should do.

FR1 – Image Capture

The system shall allow farmers to capture images of Nai Miris plant leaves using a smartphone camera.

FR2 – Image Upload

The system shall allow users to upload captured images to the cloud-based AI server for analysis.

FR3 – Image Preprocessing

The system shall preprocess uploaded images by resizing, normalizing, and enhancing image quality.

FR4 – Pest Detection

The system shall identify pest infestations in Nai Miris leaves using deep learning models such as YOLO or CNN.

FR5 – Pest Classification

The system shall classify the detected pest type affecting the plant.

FR6 – Severity Estimation

The system shall estimate the severity level of pest infestation (Low, Medium, High).

FR7 – Treatment Recommendation

The system shall provide pesticide treatment recommendations based on pest type and severity level.

FR8 – Result Display

The system shall display pest detection results, severity levels, and recommended pesticide dosage to the user.

FR9 – Data Storage

The system shall store pest detection results and image data for future analysis and research.

Non-Functional Requirements

Non-functional requirements define the system performance and quality attributes.

Security

The system should ensure secure data transmission between the mobile application and the cloud server using encryption protocols.

Performance

The system should analyze and provide pest detection results within a few seconds after the image is uploaded.

Usability

The system interface should be simple and user-friendly so that farmers with limited technical knowledge can use it easily.

Scalability

The system should support multiple users and large image datasets through cloud-based infrastructure.

Reliability

The system should provide consistent and accurate pest detection results under different environmental conditions.

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

The target market for the proposed system includes farmers in the wet zone of Sri Lanka who cultivate Nai Miris and other types of chilli varieties.

Small agricultural investors who own small-scale farming fields can be another potential target market for this system.

A customer persona for this system includes farmers who are in the age group of 25-55 years and own or operate a small agricultural field and use smartphones for accessing information.

Value Proposition

The advantages of the proposed system are as follows:

Detection of pest infestation

Estimation of severity

Prevents overuse of pesticides

Improves crop productivity

Facilitates farmers through a mobile application

With the help of this system, farmers can avoid crop loss and make their farming more efficient.

Revenue Model

The system has the potential of generating revenue through the following ways:

Subscription model of mobile application for farmers

Agricultural advisory service for commercial farms

Data analytics service for agricultural organizations

Partnering with agricultural supply companies

Cost Estimation

The costs associated with the commercialization of the system include:

Cost of developing and maintaining the AI model

Cost of cloud infrastructure

Cost of developing the mobile application

Cost of marketing and farmer training programs

Pricing Strategy

The system can be made to work on a cost-effective subscription-based pricing model, which can be feasible for small farmers.

Example of Pricing Strategy:

Free Version: Limited pest detection functionality

Premium Version: Severity estimation, treatment suggestions, etc.

Competitive Advantage

The proposed system has the following advantages over other pest detection systems:

The system has been specifically designed for crops like Nai Miris, which are commonly grown in Sri Lanka.

Severity estimation, as well as pesticide dosage suggestions, are provided.

The system utilizes AI-based image processing, which enables faster pest detection.

The system has a mobile-based interface, making it accessible to farmers.

Intellectual Property Considerations

The intellectual property rights of the developed AI models, datasets, and software architecture can be protected as follows:

Software copyrights

Ownership of the datasets created

Possibility of patenting the pest detection methodology

In order to ensure the commercialization of the system, the intellectual property rights have to be protected.

7. BUDGET AND JUSTIFICATION

Item	Description	Estimated Cost (LKR)
Hardware	Laptop with GPU capability for AI training	250,000
Mobile Device	Smartphone for image testing and data collection	60,000
Cloud Services	Google Cloud / AWS GPU training and storage	80,000
Software Tools	AI development tools, dataset annotation tools	30,000
Dataset Collection	Field visits and data collection expenses	40,000
Human Effort	Development and testing work	100,000
Total		560,000 LKR

Figure 3 - Budget

Justification

Hardware

The laptop should be high-performance for the purpose of training deep learning models and image processing.

Mobile Device

The smartphone is required for app testing and image capture in the field.

Cloud Services

Cloud computing services are required for GPU support in AI model training and data storage.

Software Tools

Software tools such as TensorFlow and PyTorch are required for AI model development.

Dataset Collection

Field visits are required for collecting real images of Nai Miris pest.

Human Effort

Development, app testing, and system implementation require human effort.

8. WORK BREAKDOWN STRUCTURE (WBS)

Phase	Task	Description
Phase 1	Problem Identification	Identify pest detection challenges in Nai Miris cultivation
Phase 2	Literature Review	Review research papers related to pest detection systems
Phase 3	Dataset Collection	Collect and label pest image datasets
Phase 4	Model Development	Develop AI models for pest detection
Phase 5	Severity Estimation	Implement pest severity classification
Phase 6	Recommendation System	Develop pesticide treatment recommendation module
Phase 7	System Integration	Combine AI modules with mobile application
Phase 8	Testing and Validation	Evaluate system accuracy and performance
Phase 9	Documentation	Prepare research report and final dissertation

Table 2- work break down

Timeline Alignment

The tasks will be allocated to different periods of the project from the research proposal to the implementation of the system.

Balanced Workload Distribution

The workload will be evenly distributed across all phases of the research to guarantee that all phases are given adequate development and testing time. This will help in efficient completion of the project within the stipulated period.

9. GANTT CHART

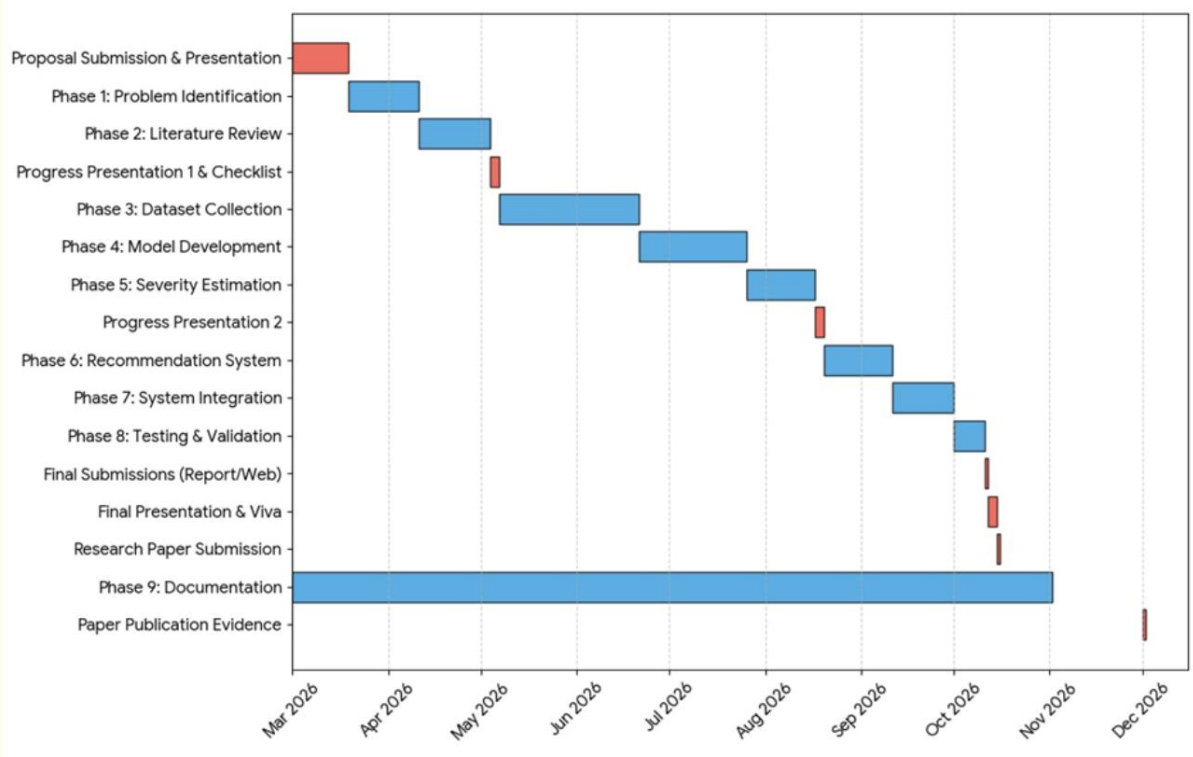


Figure 4 - GANTT CHART

10.REFERENCES LIST

[1] S. P. Mohanty, D. P. Hughes and M. Salathé, "Using Deep Learning for Image-Based Plant Disease Detection," *Frontiers in Plant Science,* vol. 7, p. 1419, 2016.

[2] K. P. Ferentinos, "Deep Learning Models for Plant Disease Detection and Diagnosis," *Computers and Electronics in Agriculture,* Vols. 311-318, p. 145, 2018.

[3] S. Sladojevic, M. Arsenovic, A. Anderla, D. Culibrk and D. Stefanovic, "Deep Neural Networks Based Recognition of Plant Diseases by Leaf Image Classification," *Computational Intelligence and Neuroscience,* vol. 2016, p. 3289801, 2016.

[4] A. Kamilaris and F. X. Prenafeta-Boldú, "Deep Learning in Agriculture: A Survey," *Computers and Electronics in Agriculture,* vol. 147, pp. 70-90, 2018.

[5] S. Zhang, X. Wu, Z. You and L. Zhang, "Leaf Image Based Cucumber Disease Recognition Using Sparse Representation Classification," *Computers and Electronics in Agriculture,* vol. 134, pp. 135-141, 2017.

[6] J. G. A. Barbedo, "A Review on the Main Challenges in Automatic Plant Disease Identification Based on Visible Range Images," *Biosystems Engineering,* vol. 144, pp. 52-60, 2016.

[7] A. Picon, A. Alvarez-Gila, M. Seitz, A. Ortiz-Barredo, J. Echazarra and A. Johannes, "Deep Convolutional Neural Networks for Mobile Capture Device-Based Crop Disease Classification in the Wild," *Computers and Electronics in Agriculture,* vol. 161, pp. 280-290, 2019.

[8] M. Brahimi, K. Boukhalfa and A. Moussaoui, "Deep Learning for Tomato Diseases: Classification and Symptoms Visualization," *Applied Artificial Intelligence,* vol. 31, no. 4, pp. 299-315, 2017.

[9] Y. Li, S. Zhang, C. Wang and B. Zhang, "Pest Detection and Recognition in Agriculture Based on Computer Vision," *Biosystems Engineering,* vol. 192, pp. 99-115, 2020.

[10] G. Wang, Y. Sun and J. Wang, "Automatic Image-Based Plant Disease Severity Estimation Using Deep Learning," *Computational Intelligence and Neuroscience,* vol. 2017, p. 2917536, 2017.

[11] A. K. Mahlein, "Plant Disease Detection by Imaging Sensors – Parallels and Specific Demands for Precision Agriculture," *Precision Agriculture,* vol. 17, no. 2, pp. 241-251, 2016.

[12] S. Sankaran, A. Mishra, R. Ehsani and C. Davis, "Review of Advanced Techniques for Detecting Plant Diseases," *Computers and Electronics in Agriculture,* vol. 72, pp. 1-13, 2010.

[13] J. G. A. Barbedo, "Digital Image Processing Techniques for Detecting, Quantifying and Classifying Plant Diseases," *SpringerPlus,* vol. 2, p. 660, 2013.

[14] C. H. Bock, G. H. Poole, P. E. Parker and T. R. Gottwald, "plant Disease Severity Estimated Visually, by Digital Photography and Image Analysis," *Critical Reviews in Plant Sciences,* vol. 29, no. 2, pp. 59-107, 2010.

[15] C. Zhang, Y. Zhou, F. Qiu and J. Wang, "Application of Deep Learning in Pest Detection and Classification for Smart Agriculture," *IEEE Access,* vol. 7, p. 146828–146837, 2019.

[16] B. Liu, Y. Zhang, D. He and Y. Li, "Identification of Apple Leaf Diseases Based on Deep Convolutional Neural Networks," *Symmetry,* vol. 10, no. 1, p. 11, 2018.

[17] J. Article, "A Robust Deep Learning-Based Detector for Real-Time Tomato Plant Diseases and Pests Recognition," *Sensors,* vol. 17, no. 9, p. 2002, 2017.

[18] Y. Lu, S. Yi, N. Zeng, Y. Liu and Y. Zhang, "Identification of Rice Diseases Using Deep Convolutional Neural Networks," *Neurocomputing,* vol. 267, p. 378–384, 2017.

[19] A.Retalis, "Geographic information systems and science," *The Photogrammetric Record,* p. 396, 2005.

[20] N. E. E.-e. El.Faouzi, "Classifiers and distance-based evidential fusion for road travel time estimation," *Proceedings of SPIE, the International Society for Optical Engineering/Proceedings of SPIE,* 2006.

[21] Clunder, G A; Bass, A P; De Beek, F J;, "A long term travel time prediction algorithm using historical data," *PROCEEDINGS OF THE 14TH WORLD CONGRESS ON INTELLIGENT TRANSPORT SYSTEMS (ITS), HELD BEIJING, OCTOBER 2007,* 2007.

[22] W. W. H. "Machine Learning Methods in the Environmental Sciences: Neural Networks and," 2009.

[23] T. Bolu, "Real-Time Data Processing in AI Applications," *ResearchGate,* 2024.

[24] Nagy, Balázs;, "A New Method of Improving the Azimuth in Mountainous Terrain by Skyline Matching," *PFG – Journal of Photogrammetry Remote Sensing and Geoinformation Science,* vol. 88, no. 2, p. 121, 2020.

[25] Fedorov, Roman; Martinenghi, Davide; Tagliasacchi, Marco; Castelletti, Andrea;, "Exploiting User Generated Content for Mountain Peak Detection," *Electronic workshops in computing,* 2013.

[26] La Salandra, Antonio; Fraternali, Piero; Frajberg, Darian;, "A Location-Based Virtual Reality Application for Mountain Peak Detection," 2018.

[27] Cam, C;, "Navigation | Map, Compass and GPS | The Hiking Life," 11 October 2021. [Online]. Available: https://www.thehikinglife.com/hiking-and-backpacking-skills/navigation/.

[28] Macdonald, Makai;, "How Weather Influences Trail Conditions for Hikers," 07 May 2024. [Online]. Available: https://www.macfishes.com/impact-of-weather-on-trail-conditions/. [Accessed 24 August 2024].

11.APPENDICES

Appendix A – System Architecture Diagram

This appendix presents the detailed system architecture diagram of the proposed AI-Driven Pest Monitoring and Control System for Nai Miris cultivation. The diagram illustrates the interaction between the image capture module, image processing module, pest detection model, pest severity analysis module, and treatment recommendation module hosted on the cloud server.

Appendix B – Literature Review Comparison Table

This appendix contains the literature review comparison table used in the research proposal. The table summarizes key previous studies related to AI-based pest detection systems, including the techniques used, datasets, accuracy results, and limitations of each approach. This comparison helps to highlight the research gap addressed by the proposed system.

Appendix C – Work Breakdown Structure (WBS)

This appendix presents the Work Breakdown Structure (WBS) for the research project. The WBS divides the project into smaller tasks such as literature review, dataset collection, model development, system implementation, evaluation, and report writing. It helps in organizing and managing the project activities effectively throughout the timeline.